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Predicting ICU Mortality with Logistic Regression and XGBoost

Introduction

Predicting in-hospital mortality for intensive care unit (ICU) patients is critical for prioritizing care, allocating resources, and informing early interventions. Traditional models (e.g., APACHE) use limited variable sets and linear assumptions, whereas modern ML methods can capture nonlinearities and interactions in heterogeneous EHR data.

We used the multi-center WiDS/MIT GOSSIS ICU dataset (first 24h features; 130k+ admissions) to compare L1-regularized logistic regression and gradient-boosted trees (XGBoost). Our clinical objective emphasized minimizing missed mortality (high recall), with threshold tuning to balance alert burden and sensitivity.

Methods

Data and Preprocessing

From first-24h variables we excluded high-missingness labs, filtered collinearity (absolute correlation > 0.80), and renamed the outcome to *Death*. Missing numeric values were imputed with means; categorical values with the most frequent level. Numeric features were standardized and categorical features one-hot encoded via a `ColumnTransformer` pipeline. We performed a 90/10 stratified split.

Models and Tuning

Baseline was L1-penalized `LogisticRegression` (solver "saga") with and without class balancing. For XGBoost we trained unweighted and class-weighted models (using `scale_pos_weight`) with randomized search (30 candidates; 3-fold CV) over maximum depth, learning rate, subsampling, column subsampling, estimators, and L1/L2 regularization. We tuned decision thresholds by maximizing F2 and by meeting a recall constraint.

Evaluation

Primary metric was AUROC; we also examined precision–recall (average precision). At selected thresholds we report accuracy, precision, recall, F1, and false-positive rate. SHAP was used for global and patient-level interpretation of the weighted XGBoost model.

Results

Baseline Characteristics (Table 1)

Table 1 summarizes baseline characteristics by in-hospital mortality status (Alive vs Death). Continuous variables are shown as mean (SD); categorical variables as counts (percentages).

Variable	Overall	Alive (0)	Death (1)
Age	62.31 (16.78)	61.75 (16.86)	68.49 (14.43)
BMI	29.19 (8.28)	29.26 (8.22)	28.34 (8.75)
Pre-ICU LOS	0.84 (2.49)	0.79 (2.36)	1.35 (3.55)
APACHE II Diagnosis	185.40 (86.05)	187.78 (86.33)	160.25 (78.72)
APACHE III Diagnosis	558.22 (463.27)	571.10 (468.88)	421.15 (372.01)
GCS Eyes	3.47 (0.95)	3.54 (0.87)	2.65 (1.31)
GCS Verbal	3.99 (1.56)	4.11 (1.48)	2.76 (1.80)
Heart Rate	99.71 (30.87)	98.69 (30.30)	110.56 (34.56)
Mean Arterial Pressure	88.02 (42.03)	88.54 (41.39)	82.46 (48.00)
Respiratory Rate	25.81 (15.11)	25.41 (15.07)	30.08 (14.81)
Temperature	36.41 (0.83)	36.45 (0.74)	35.98 (1.46)
BUN	25.83 (20.67)	24.65 (19.87)	37.81 (24.56)
Creatinine	1.48 (1.53)	1.43 (1.50)	2.04 (1.66)
Glucose	160.33 (90.79)	158.46 (89.03)	179.78 (105.50)
Hematocrit	32.99 (6.87)	33.12 (6.79)	31.61 (7.55)
Sodium	137.97 (5.28)	137.94 (5.10)	138.19 (6.83)
WBC	12.13 (6.92)	11.84 (6.53)	15.12 (9.57)

Variable	Overall	Alive (0)	Death (1)
Elective Surgery: Yes	16851 (18.4%)	16330 (19.5%)	521 (6.6%)
Acute Renal Failure: Yes	2546 (2.8%)	2212 (2.7%)	334 (4.3%)
Ventilated: Yes	29640 (32.6%)	24355 (29.3%)	5285 (67.5%)
AIDS: Yes	78 (0.1%)	68 (0.1%)	10 (0.1%)
Cirrhosis: Yes	1428 (1.6%)	1180 (1.4%)	248 (3.2%)
Diabetes: Yes	20492 (22.5%)	18897 (22.7%)	1595 (20.4%)
Hepatic Failure: Yes	1182 (1.3%)	968 (1.2%)	214 (2.7%)
Immunosuppression: Yes	2381 (2.6%)	1997 (2.4%)	384 (4.9%)
Leukemia: Yes	643 (0.7%)	524 (0.6%)	119 (1.5%)
Lymphoma: Yes	376 (0.4%)	313 (0.4%)	63 (0.8%)
Solid Tumor Metastasis: Yes	1878 (2.1%)	1531 (1.8%)	347 (4.4%)
Gender: F	42219 (46.0%)	38488 (45.9%)	3731 (47.2%)
Gender: M	49469 (54.0%)	45293 (54.1%)	4176 (52.8%)
Ethnicity: African American	9547 (10.6%)	8797 (10.7%)	750 (9.6%)
Ethnicity: Asian	1129 (1.3%)	1036 (1.3%)	93 (1.2%)
Ethnicity: Caucasian	70684 (78.3%)	64516 (78.2%)	6168 (79.0%)
Ethnicity: Hispanic	3796 (4.2%)	3420 (4.1%)	376 (4.8%)
Ethnicity: Native American	788 (0.9%)	718 (0.9%)	70 (0.9%)
Ethnicity: Other/Unknown	4374 (4.8%)	4021 (4.9%)	353 (4.5%)
ICU Admit Source: Accident & Emergency	54060 (59.0%)	49390 (59.0%)	4670 (59.1%)
ICU Admit Source: Floor	15611 (17.0%)	13517 (16.1%)	2094 (26.5%)
ICU Admit Source: Operating Room / Recovery	18713 (20.4%)	18015 (21.5%)	698 (8.8%)
ICU Admit Source: Other Hospital	2358 (2.6%)	2041 (2.4%)	317 (4.0%)
ICU Admit Source: Other ICU	859 (0.9%)	735 (0.9%)	124 (1.6%)

Variable	Overall	Alive (0)	Death (1)
ICU Type: Med-Surg ICU	50586 (55.2%)	46160 (55.1%)	4426 (55.9%)
ICU Type: Other ICU types	41127 (44.8%)	37638 (44.9%)	3489 (44.1%)

Continuous variables are mean (SD); categorical variables are n (%).

[Download Table 1 \(CSV\)](#)

Non-survivors were older and more acutely ill at presentation: mean age was **68.5** years in the Death group versus **61.8** years in the Alive group, and pre-ICU length of stay was longer (**1.35** vs **0.79** days). Neurologic status was worse among those who died (GCS Eyes **2.65** vs **3.54**; GCS Verbal **2.76** vs **4.11**). Vital signs showed higher heart rate (**110.6** vs **98.7** bpm) and respiratory rate (**30.1** vs **25.4** breaths/min), lower mean arterial pressure (**82.5** vs **88.5** mmHg), and lower temperature (**36.0** vs **36.5** °C) among non-survivors. Markers of organ dysfunction and inflammation were elevated in deaths (BUN **37.8** vs **24.7** mg/dL; creatinine **2.04** vs **1.43** mg/dL; WBC **15.1** vs **11.8** ×10⁹/L). Mechanical ventilation was far more common in deaths (**67.5%** vs **29.3%**), while elective surgery was less frequent (**6.6%** vs **19.5%**). Deaths were more often admitted from the floor (**26.5%** vs **16.1%**) and less often from the operating room/recovery (**8.8%** vs **21.5%**), and occurred predominantly in medical-surgical ICUs (**55.9%** vs **55.1%**), mirroring the overall ICU case-mix.

Logistic Regression

Unweighted LASSO achieved AUROC 0.86 and accuracy 0.92. Class 0 (Alive) precision, recall, and F1 were 0.93, 0.99, and 0.96; Class 1 (Death) precision, recall, and F1 were 0.61, 0.16, and 0.26. With class-balanced weights, AUROC remained 0.86; accuracy decreased to 0.78 while Death recall improved to 0.77 (precision 0.25; F1 0.38).

XGBoost

The unweighted model (maximum depth five; learning rate 0.10; subsampling 1.0; column subsampling 0.60; 200 estimators; L1 0.10; L2 1) produced AUROC 0.895 and accuracy 0.93. Death precision and recall were 0.69 and 0.29 (F1 0.41). The weighted model (maximum depth four; learning rate 0.10; subsampling 1.0; column subsampling 0.80; 200 estimators; L1 1; L2 1) achieved AUROC 0.891 with Death recall 0.79 (precision 0.28; accuracy 0.81; F1 0.42).

To visualize operating characteristics, we compared receiver operating characteristic (ROC) and precision–recall (PR) curves for weighted XGBoost and weighted LASSO on the holdout. ROC curves summarize discrimination across thresholds, whereas PR curves emphasize performance on the minority class by plotting precision against recall.

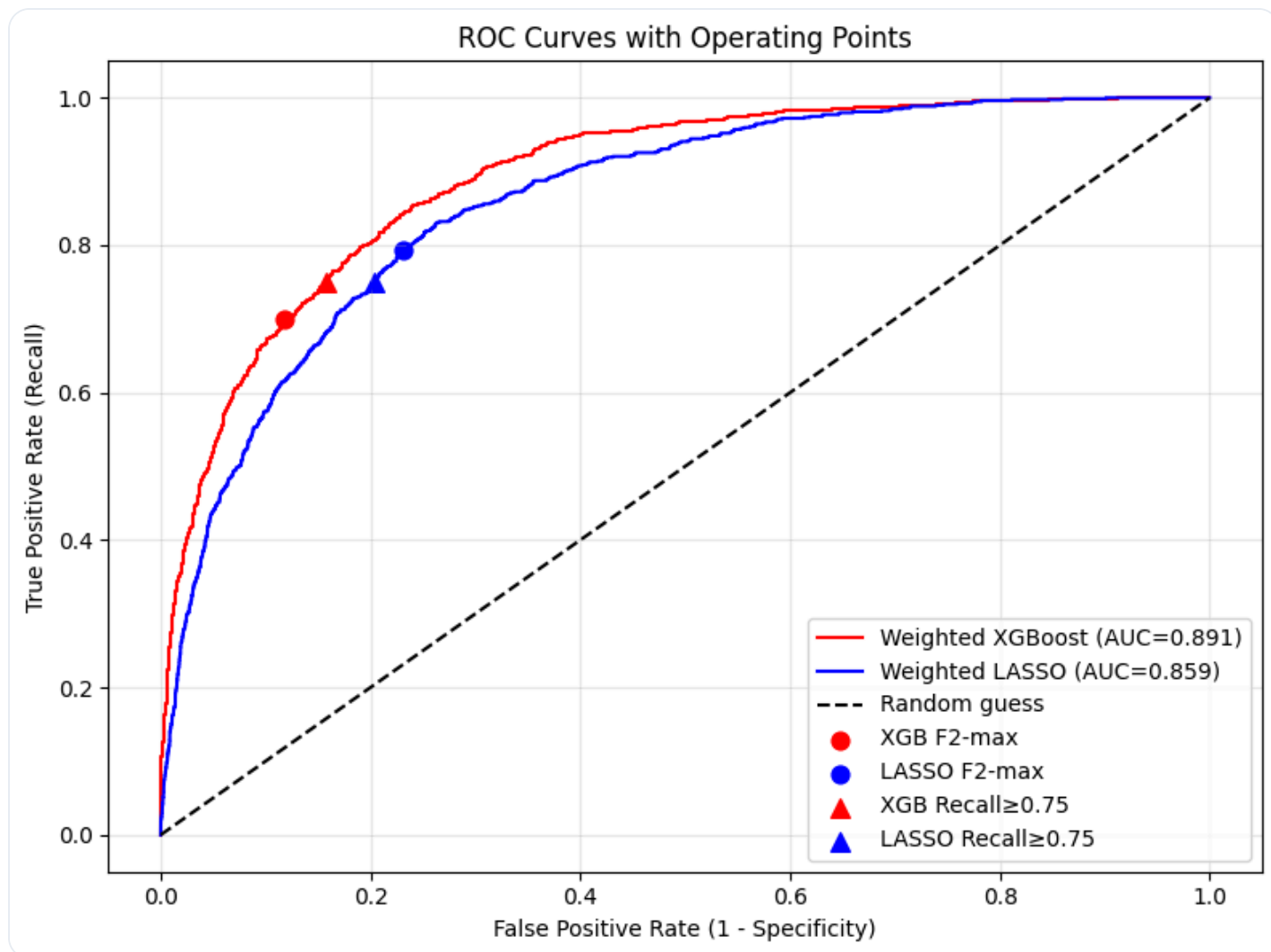


Figure A. ROC curves comparing weighted XGBoost and weighted LASSO logistic regression (AUC shown in legend).

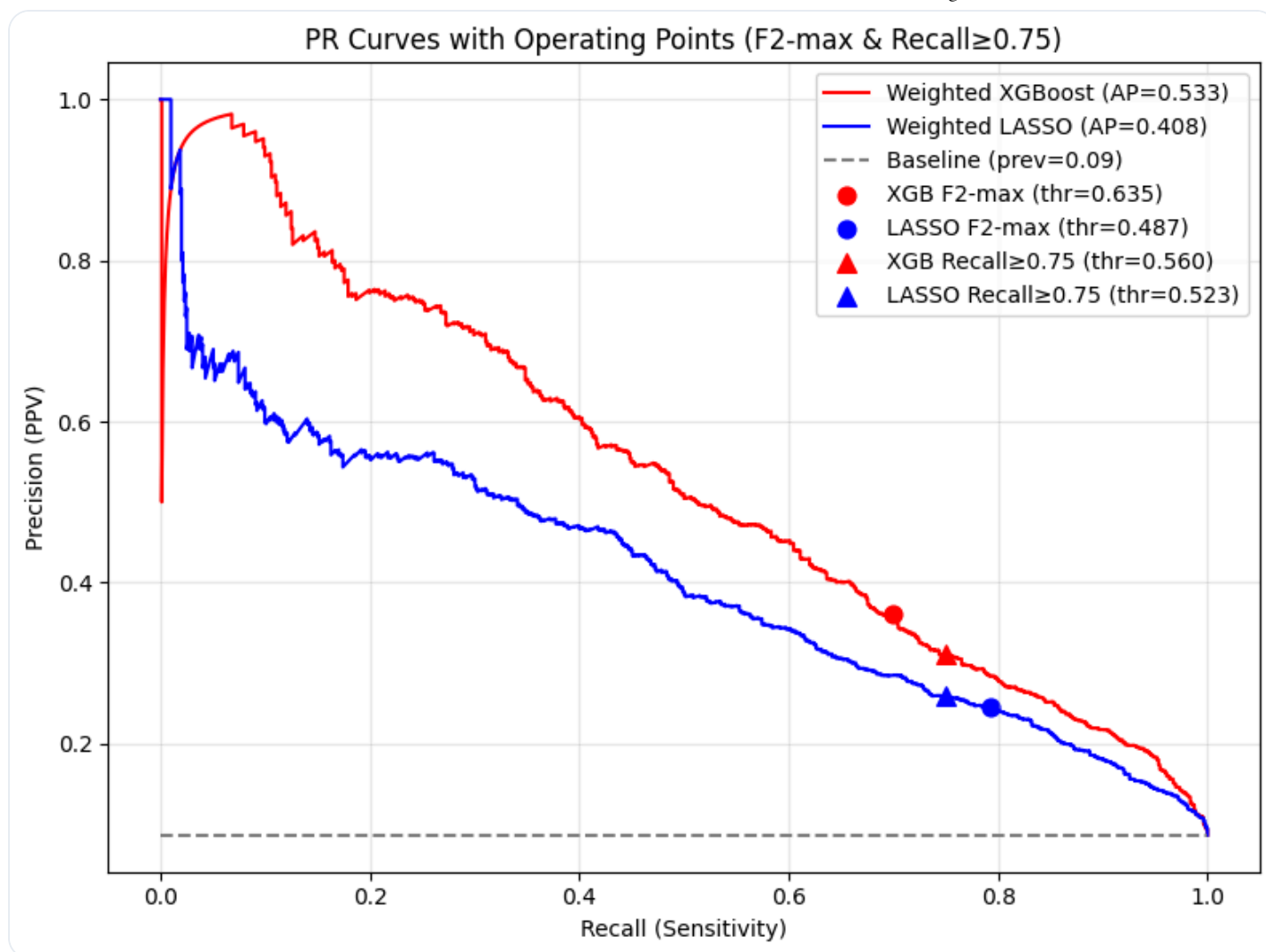


Figure B. Precision–Recall curves comparing weighted XGBoost and weighted LASSO; the dashed line marks the prevalence baseline.

Consistent with the numeric results, the ROC curves show comparable overall discrimination, while the PR curves highlight that, at recall near 0.75, weighted XGBoost maintains higher precision (fewer false alarms) than weighted LASSO. This supports selecting weighted XGBoost when alert burden must be controlled under a fixed sensitivity target.

Model Interpretation (SHAP)

Global SHAP analysis highlighted ventilation status, age, Glasgow Coma Scale subscores, heart rate, mean arterial pressure, respiratory rate, and temperature as major drivers of risk. Renal and inflammatory markers (BUN, creatinine, WBC), APACHE II/III diagnoses, pre-ICU length of stay, and elective surgery also contributed. A representative waterfall plot showed how abnormal temperature, reduced GCS, ventilation, tachycardia, and hypotension pushed a prediction toward death, while younger age and lower creatinine were protective.

We display the SHAP summary plot to visualize global importance and directionality across all patients, followed by a waterfall plot for a representative stay to illustrate feature-wise contributions for an individual prediction.

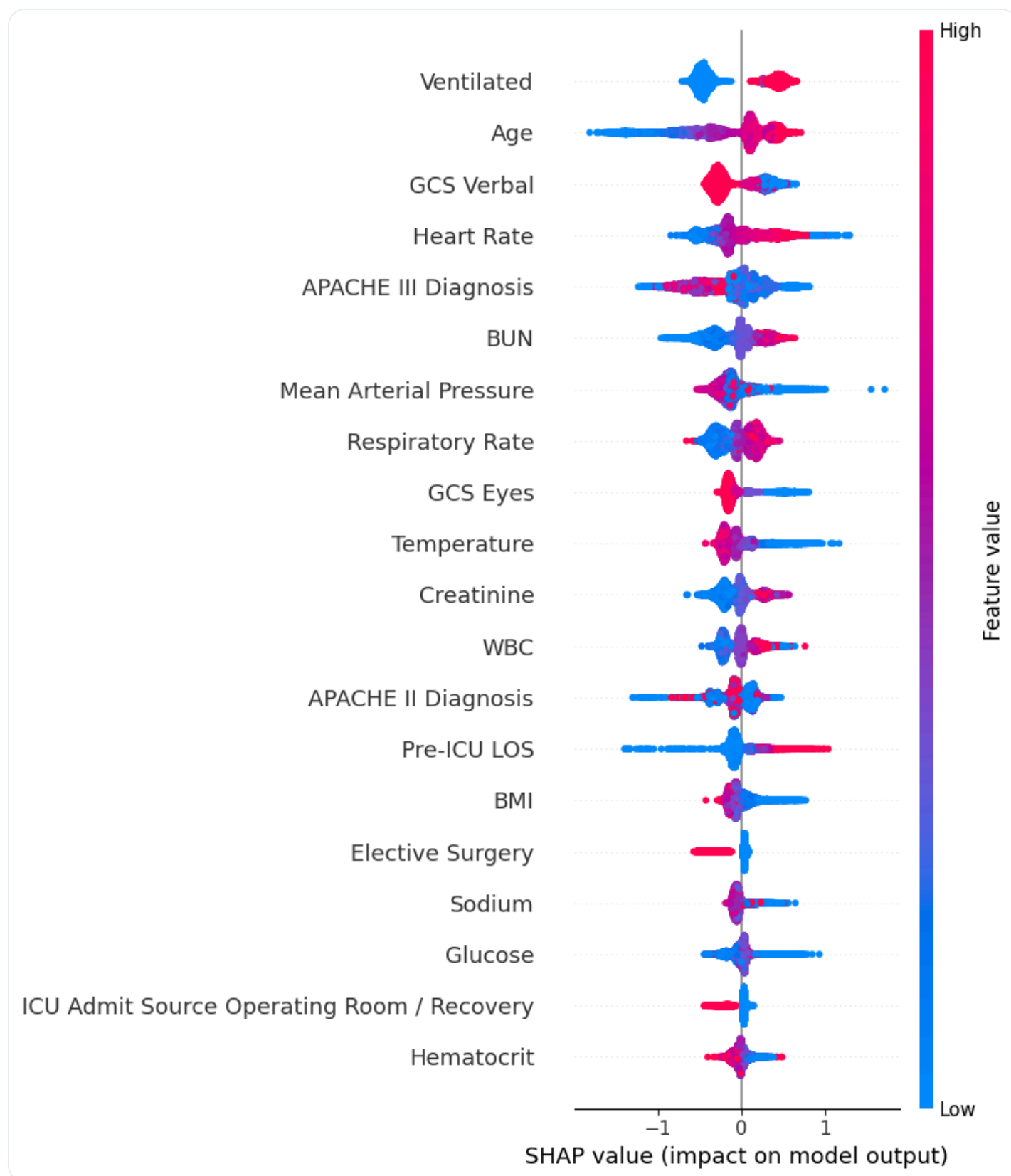


Figure C. SHAP summary plot (weighted XGBoost). Each point is a patient; color encodes feature value (blue = low, red = high). Higher SHAP values increase predicted mortality.

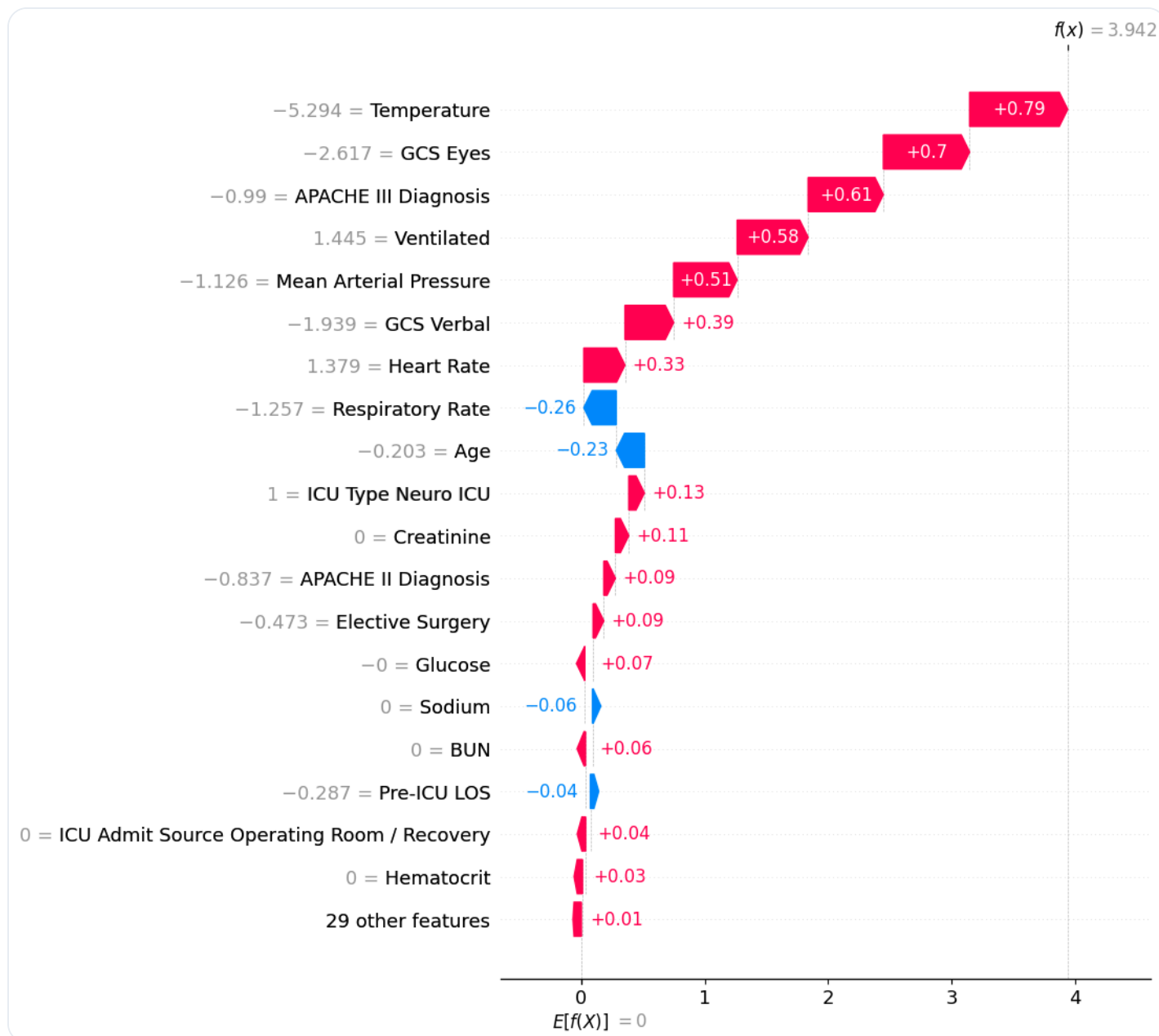


Figure D. SHAP waterfall for a representative ICU stay, showing feature-wise contributions from the baseline toward the predicted log-odds of death.

These plots confirm that the model's risk signals align with known pathophysiology—respiratory failure, neurologic depression, hemodynamic instability, and renal dysfunction—supporting face validity and enabling transparent, patient-level rationale for risk scores in clinical review.

Discussion

Both models demonstrated strong discrimination. Weighting strategies substantially improved sensitivity for mortality, consistent with our objective of minimizing false negatives. At matched sensitivity (recall ≈ 0.75), weighted XGBoost provided higher precision and a lower false-positive rate than weighted LASSO, suggesting a more favorable alert burden when

resources are limited; conversely, the F2-optimal LASSO setting may suit contexts prioritizing maximal capture of deaths despite more false positives.

SHAP explanations aligned with established risk factors (ventilation, age, neurologic status, hemodynamics, organ dysfunction), supporting face validity and enabling transparent, patient-level rationale. Limitations include use of first-24h features only, simple imputation, potential miscalibration with weighting, and internal validation only. Future work should examine calibration and decision-curve analyses, subgroup fairness, temporal trajectories, and external prospective validation with drift monitoring.

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